

GANESH REDDY

Erlangen, Germany

+49 1551 0182395 — ganeshreddy30102000@gmail.com — GitHub — LinkedIn

Profile

Master's student in **Autonomy Technologies** with strong focus on **medical image analysis, computer vision, and deep learning**. Hands-on experience in **chest X-ray-based disease detection, CNN training in low-data regimes, and transfer learning using public medical imaging datasets**. Proficient in **Python, PyTorch, and OpenCV**, with experience training supervised regression and classification models. Highly motivated to work in a **clinical hospital environment** and investigate domain shift, data scarcity, and generalization challenges in medical imaging.

Actively seeking a Master's thesis and team project in medical image computing, low-data learning, and clinical AI.

Education

Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU)

M.Sc. Autonomy Technologies

ECTS: 62.5 / 120 Current GPA: 2.7

Erlangen, Germany

Apr 2024 – Present

Relevant Coursework:

- Introduction to Deep Learning
- Pattern Recognition
- Computer Vision
- Machine Learning in Signal Processing
- AI in Medical Robotics
- Robotics in Surgery and Diagnostics
- Human-Robot Co-adaptation

KLE Technological University

B.Tech Automation and Robotics

GPA: 2.3

Bachelor's Thesis: Automated Grading and Sorting System using Computer Vision

Hubballi, India

Aug 2018 – Jul 2022

Research & Academic Experience

Student Research Assistant (Part-time) - IDEA LABS

Friedrich-Alexander-Universität Erlangen–Nürnberg (FAU)

Nov 2025 – Present

- Develop and maintain Python-based data analysis pipelines for scientific datasets
- Perform data preprocessing, validation, and quality checks on structured datasets
- Support debugging, reproducibility, and evaluation of machine learning workflows
- Create automated evaluation scripts and structured experiment reports

Medical Imaging & Computer Vision Projects

Lung Disease Detection from Chest X-ray Images

- Trained CNN-based models for pneumonia and tuberculosis classification using public chest X-ray datasets
- Applied transfer learning and data augmentation to improve performance in low-data settings
- Implemented preprocessing pipelines including normalization, resizing, and augmentation
- Evaluated model robustness and generalization across patient distributions

Medical Image Classification using Deep Learning

- Curated and cleaned medical imaging datasets with focus on data consistency and bias
- Trained supervised learning models and analyzed failure cases
- Identified common pitfalls in medical imaging pipelines related to dataset quality

Technical & Industry Experience

Internet of Things (IoT) Intern

Jan 2023 – Oct 2023

Digital Associates India, Hyderabad

- Designed Python-based data processing pipelines for large real-world sensor datasets
- Implemented OpenCV-based computer vision pipelines for detection and validation tasks
- Performed data cleaning, validation, and structured reporting in Agile workflows

Systems Engineer Intern

Jan 2022 – Jul 2022

Centre for Autonomous Systems Engineering, Hubballi

- Developed robotics software using ROS2 (Python/C++) and Gazebo simulation
- Designed and validated kinematic models for a 5-DOF robotic arm
- Conducted systematic testing and performance evaluation through simulation

Technical Skills

Programming: Python, C++, MATLAB

Deep Learning & ML: PyTorch, TensorFlow, CNNs, supervised regression and classification

Medical Imaging & Vision: Chest X-ray analysis, OpenCV, image preprocessing, dataset curation

Tools: Linux, Git, Docker (basic), VS Code

Research Interests

Medical Image Computing, Low-Data Deep Learning, Domain Adaptation, Foundation Models for Medical Imaging, Clinical AI Systems

Languages

English (C1) German (A2) Hindi (Native)

Availability

Available for **Master's Thesis and Team Project**. Open to working full-time in hospital-based research environments with limited compute resources.